Photocatalytic Degradation of Ciprofloxacin Using a $BiFeO_3/TiO_2$ Heterostructure with Peroxymonosulfate Activation(PMS)

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Abstract

The $BiFeO_3/TiO_2$ heterojunction was evaluated for ciprofloxacin degradation in the presence of PMS. The system showed high efficiency under both UV and visible light due to improved charge separation and extended light absorption. Reactive species ($SO_4^{\bullet -}$ and •OH) played a key role in degradation. The results highlight the composite as a promising photocatalyst for water treatment.

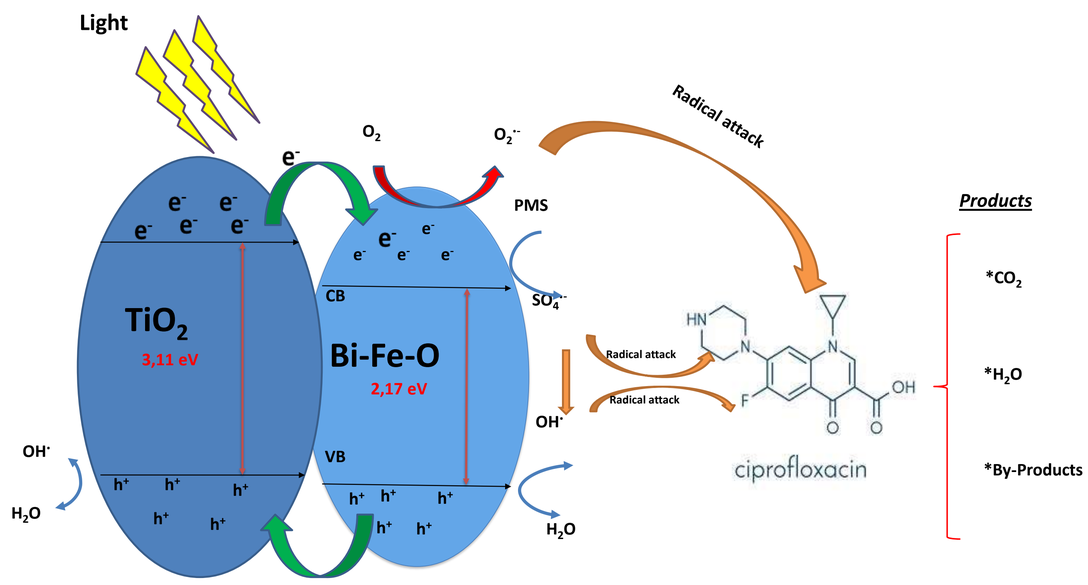
Introduction

The presence of pharmaceutical contaminants such as ciprofloxacin in water systems has raised serious environmental concerns due to their persistence and resistance to conventional treatment methods. Advanced oxidation processes (AOPs), particularly photocatalysis, have emerged as effective approaches for degrading such pollutants.

Among photocatalysts, TiO_2 is widely used due to its stability and strong oxidative ability; however, its wide band gap limits its activity to UV light. To overcome this limitation, coupling TiO_2 with narrow band gap semiconductors such as $BiFeO_3$ has gained attention. $BiFeO_3$ not only extends light absorption into the visible region but also promotes efficient charge separation through heterojunction formation.

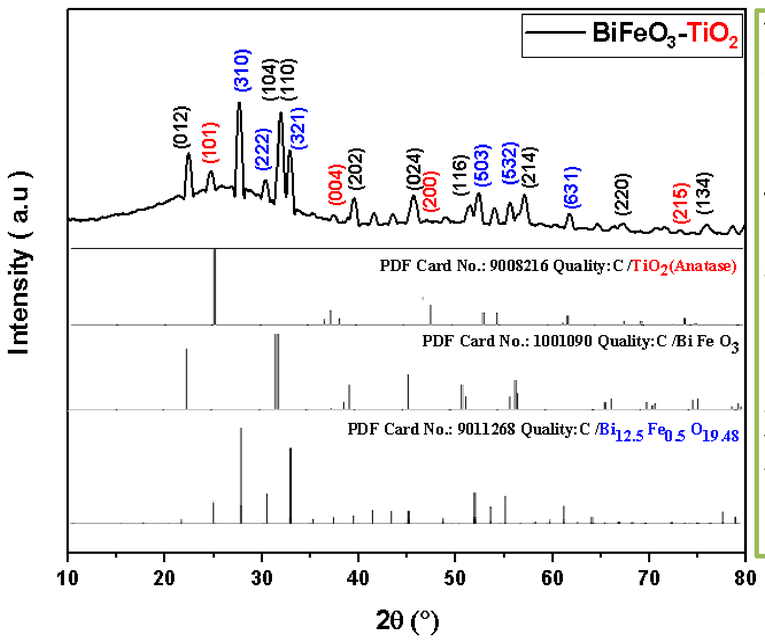
In addition, the incorporation of PMS enhances the system by generating highly reactive radicals, further improving degradation efficiency. This study focuses on the synthesis, characterization, and photocatalytic performance of a $BiFeO_3/TiO_2$ composite for CIP degradation under different irradiation conditions.

Proposed Mechanism



EXPERIMENTAL STUDY

• Structural Analysis XRD



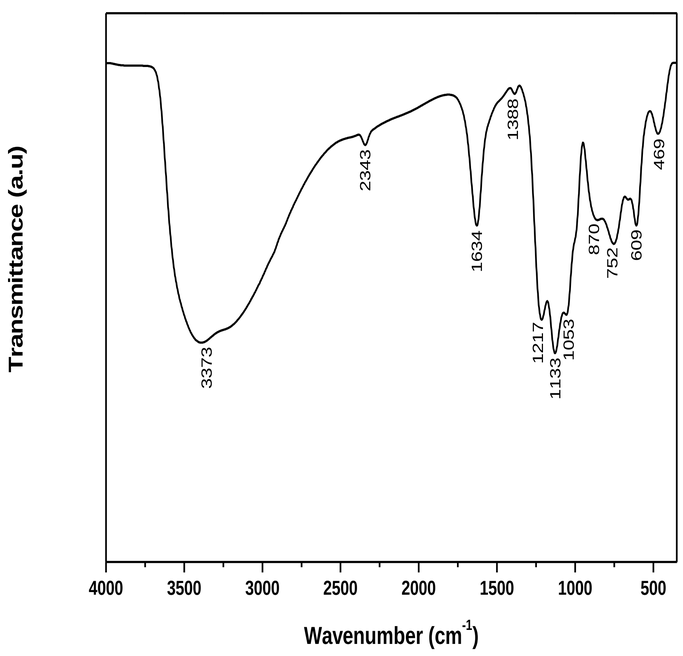
The XRD pattern confirms the successful synthesis of a multi-phase heterostructure:

Primary Phases:
Clear identification of **Anatase TiO_2** (25.3°, 37.8°) and **Perovskite $BiFeO_3$** (31°, 40°).

Secondary Phase:
Presence of **Sillénite-type $Bi_{12.5}Fe_{0.5}O_{19.48}$**, a common Bi-rich byproduct.

Synergy: The interaction between these three phases promotes visible-light absorption and efficient charge separation

• Fourier Transform Infrared (FTIR) spectroscopy



The FTIR spectrum confirms the successful formation of the nanocomposite:

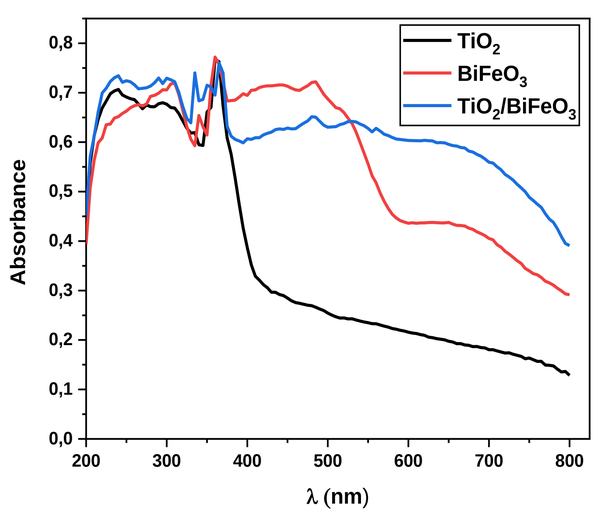
Perovskite $BiFeO_3$: Peak at **609 cm^{-1}** (Fe–O stretching).

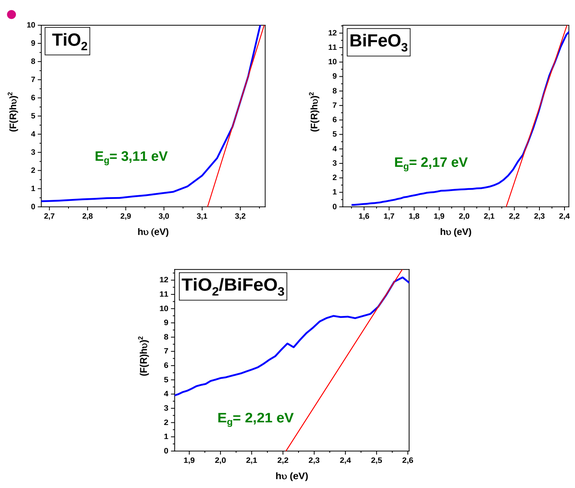
Anatase TiO_2: Band at **469 cm^{-1}** (Ti–O–Ti vibrations).

Secondary Phase: Sharp peaks at **752** and **870 cm^{-1}** indicate a secondary Bi-Fe-O phase.

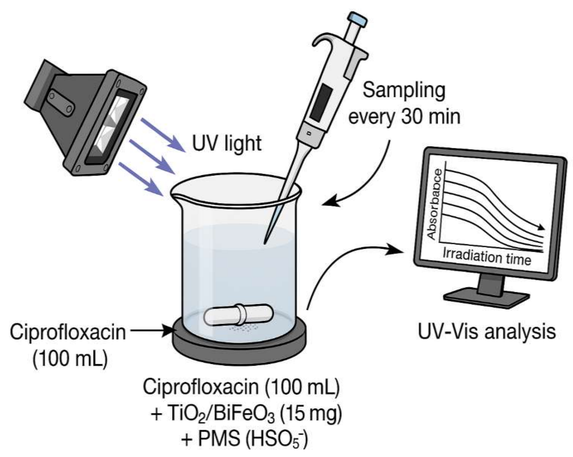
Surface Groups: Large bands at **3000—3600 cm^{-1}** and **1634 cm^{-1}** represent adsorbed hydroxyls and water.

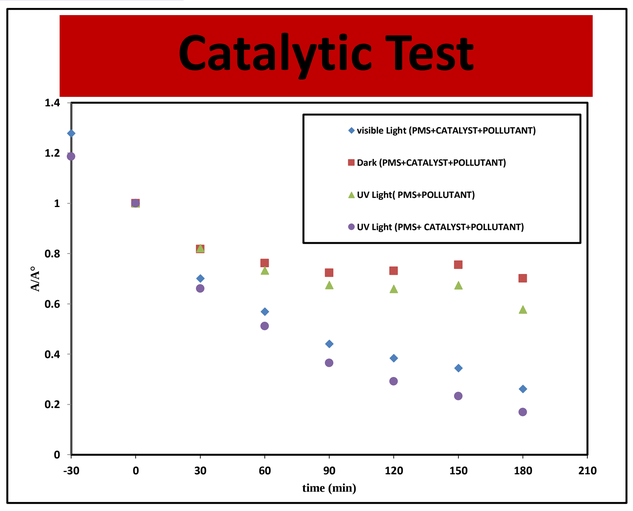
• Optical Properties-UV-Vis DRS





• Photocatalytic Degradation of CIP





Conclusion and perspective

The $BiFeO_3/TiO_2$ system showed enhanced photocatalytic performance due to synergistic effects and efficient radical generation. It is effective under both UV and visible light, making it suitable for wastewater treatment.

Future work should focus on improving stability, optimizing phase composition, and applying the system in real wastewater under solar light.